

Chapter 17 — Verification and Simulation

Field	Value
Status	Revised after whole-book constitutional and cross-chapter audit
Version	0.3
Scope	Accumulate risk-proportionate evidence before changing reality and continuously verify critical recovery paths.
Constitutional basis	Decisions 28, 35, 43, 47, 50–51, 55–56
Owner review	No unresolved Owner decision after whole-book audit

1. Purpose

Accumulate risk-proportionate evidence before changing reality and continuously verify critical recovery paths.

2. Authoritative Responsibilities

Component	Authoritative responsibility
Verification Service	Verification Plans, Records, thresholds and disposition
Simulation Service	Isolated models and simulated environments
Digital Twin	Living model of topology, services, policies, trust, resources, intent and failure modes
Reality Boundary Controller	Simulation, shadow, test, canary, production, recovery and emergency separation

3. Core State and Records

Object	Required content
Verification Plan	Risk class, applicable levels, evidence required, budget, authority and stopping rules.
Verification Record	Inputs, environment, tests, results, fidelity, confidence, signatures and reproducibility.
Simulation Fidelity	Coverage, accuracy, missing variables, limitations and historical predictive success.
Canary/Shadow Observation	Scope, comparison baseline, drift, health and rollback trigger.

4. Runtime Workflow

1. Classify the proposed change by consequence and reversibility.
2. Run static validation for syntax, permissions, dependencies, signatures and compatibility.
3. Use simulation and Digital Twin scenarios; score fidelity rather than treating all simulations equally.
4. Inject bounded failures and evaluate counterfactual alternatives inside the isolated Reality Boundary.
5. Run shadow execution without control and compare predictions with reality.
6. Deploy canary or controlled reality with bounded blast radius and automatic rollback where valid.
7. Adopt fully only after evidence and required Owner authorization; continue post-change verification.

5. Interfaces and Contracts

All commands, queries and events are typed, versioned, authenticated and correlated. Commands request mutation from the authoritative owner; queries declare consistency and privacy scope; immutable events record completed facts with identity, causation, authorization, provenance and integrity metadata.

6. Failure and Recovery

- Simulation leaks toward reality: Reality Boundary blocks the effect and raises a constitutional incident.
- Low fidelity: reduce confidence and require stronger real-world evidence.
- Canary degradation: stop expansion and rollback or recover according to transaction policy.
- Verification service unavailable: protected or high-risk change does not proceed.

7. Constitutional Guarantees

- HAL earns confidence through verification before changing reality.
- Simulation informs authority but never replaces it.
- Every important verification is reproducible and permanently recorded.
- Verification burden scales with risk; urgency cannot remove constitutional authorization.

8. Prohibited Behaviors

- No component may silently broaden authority, reinterpret Owner intent, erase dissent, or treat a derived projection as authoritative state.
- No degraded dependency may silently change policy, identity, trust, authentication, verification or evidence requirements.
- No provider, model, node, Presence, secret, relationship or network location receives constitutional authority by implication.

9. Observability and Verification

The subsystem records correlation, identity, policy/authority context, state versions, decisions, limitations, failures, recovery and outcomes. Verification is risk-scaled and produces durable evidence. Status reporting is derived from governed state rather than model assertion.

10. Source Alignment and Review

This chapter directly implements Decisions 28, 35, 43, 47, 50–51, 55–56. The v0.2 rewrite restores the source-specific objects, workflows and guarantees missing from v0.1. Final disposition is recorded in the separate source-alignment and constitutional-audit reports.

Whole-Book Audit Resolution

The following requirements supersede any ambiguous earlier wording in this chapter and are incorporated into this edition:

- Embedded author approval is not conformance evidence. This chapter's constitutional status is determined only by the independent whole-book audit.
- No substantive constitutional or cross-chapter correction was required in this edition.